Machine Learning Study Notes - Part 1: Training Basics

Overfitting happens when a model memorizes the training data instead of learning the underlying pattern. A typical sign is high accuracy on the training set but much lower accuracy on the test set. Common remedies are collecting more data, using a simpler model, applying regularization, and using early stopping.

Gradient descent is the optimization algorithm that trains most models. It computes the gradient of the loss with respect to each weight and moves the weights in the opposite direction. The learning rate controls the size of each step. If the learning rate is too large the loss can diverge, and if it is too small training becomes very slow.

A train test split keeps part of the data hidden from the model so we can measure how well it generalizes. Cross validation repeats the split several times, typically five folds, and averages the scores to give a more reliable estimate.

Precision is the fraction of predicted positives that are truly positive. Recall is the fraction of actual positives that the model finds. The F1 score is the harmonic mean of precision and recall. Accuracy alone is misleading when the classes are imbalanced.

Machine Learning Study Notes - Part 2: Deep Learning

A neural network is a stack of layers. Each layer computes a weighted sum of its inputs plus a bias and then applies a nonlinear activation function such as ReLU. Without activation functions the whole network collapses into a single linear model.

Convolutional neural networks use small filters that slide over an image to detect edges, textures, and shapes. The same filter weights are shared across the whole image, which reduces the number of parameters. Max pooling shrinks the feature maps and makes the model less sensitive to small shifts.

Transfer learning reuses a model pretrained on a large dataset such as ImageNet. We freeze the early layers, replace the final classification layer with one that matches our number of classes, and train only the new layer. This works well when we have little data.

The PyTorch training loop has five steps: zero the gradients, run the forward pass, compute the loss, call backward to compute gradients, and call the optimizer step to update the weights. Forgetting to zero the gradients makes them accumulate across iterations.

Machine Learning Study Notes - Part 3: Transformers and LLMs

The transformer architecture is built around attention. Every token produces a query, a key, and a value vector. The query of one token is compared with the keys of all tokens to produce attention weights, and the output is the weighted sum of the values. Transformers process all tokens in parallel, which makes training faster than recurrent networks.

Tokenization splits text into subword units called tokens, and each token is mapped to an integer id and then to an embedding vector. Arabic text usually produces more tokens than English text of the same meaning, which increases the cost of using an API.

A hallucination is a fluent but incorrect statement produced by a language model. It happens because the model predicts the most likely next token and has no built in mechanism to verify facts.

Retrieval augmented generation, or RAG, reduces hallucinations. The system first retrieves the passages most relevant to the question from a document collection, then places them in the prompt and instructs the model to answer only from that context. Good RAG systems also return the source file and page number so answers can be verified.

Evaluation is essential. For RAG we measure retrieval quality separately, by checking whether the correct passage appears in the top results, and generation quality separately, by checking whether the final answer is correct and grounded in the sources.